

D315

Network and Security

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A network technician at a 40-person law firm replaces an old hub with a device that learns MAC addresses and forwards each frame only to the port where the destination device is connected. The new device came from a local reseller and has blue status lights. At which OSI layer does it mainly operate?

- A. Layer 1, physical, because it has ports and status lights.
- B. Layer 4, transport.
- C. Layer 2, data link.
- D. Layer 3, network.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

Which device makes forwarding decisions using IP addresses at Layer 3?

- A. Hub.
- B. Router.
- C. Switch.
- D. Wireless access point.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

A company's video-conference system streams live audio and video to remote staff. The vendor charges per seat, and IT runs it over the main office's fiber connection. Engineers want the lowest delay and can accept an occasional dropped frame. Which transport protocol fits best?

- A. TCP, because it guarantees every packet arrives.
- B. UDP.
- C. TCP, because fiber links require it.
- D. ICMP.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

An administrator must let remote staff reach an internal Linux server through an encrypted command-line session. The server has 64 GB of RAM and was installed last spring. Which port should the firewall allow?

- A. TCP 22.
- B. TCP 64.
- C. TCP 3389.
- D. TCP 23.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

A branch office is assigned the network 192.168.10.0/26. The branch has 45 employees and two printers. How many usable host addresses does the subnet provide?

- A. 64
- B. 47
- C. 254
- D. 62

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6. [TRANSFER — reversed stem / different variable]

A team needs one subnet with at least 100 usable host addresses, using the smallest block that works. Which prefix length should it use?

- A. /32
- B. /26
- C. /25
- D. /24

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

A hospital's patient portal went offline for six hours after a flood of traffic from thousands of compromised devices. No records were viewed or changed. The portal was built by an outside vendor in 2019. Which part of the CIA triad was most directly affected?

- A. Confidentiality.
- B. Availability.
- C. Integrity.
- D. Integrity, since vendor code from 2019 may be outdated.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8. [TRANSFER — reversed stem / different variable]

Which control mainly protects integrity?

- A. A firewall, because it protects all three equally.
- B. Installing redundant power supplies.
- C. Hashing files and comparing hash values to detect changes.
- D. Encrypting data at rest.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

Two offices must send a large volume of backup data to each other every night, quickly, and they have already exchanged a secret key securely. The data is about 2 TB and the link is 1 Gbps. Which approach is most efficient for encrypting the bulk data?

- A. Symmetric encryption such as AES.
- B. No encryption, because a 1 Gbps link is private enough.
- C. Asymmetric encryption such as RSA for all of the data.
- D. Hashing with SHA-256.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

A new payroll clerk at a 300-employee manufacturer needs to enter hours and run pay reports. Her manager has admin rights to the whole HR system and suggests copying his permissions to save time. The clerk transferred from the company's Chicago plant. What should the administrator do?

- A.** Give her read-only access to the entire HR system.
- B.** Copy the manager's admin permissions.
- C.** Grant full admin because she is an experienced transfer.
- D.** Grant only the permissions needed for payroll entry and pay reports.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A retailer's leaders decide that after an outage the online store must be running again within 4 hours, and no more than 15 minutes of orders may be lost. The store runs in two cloud regions and processes 1,200 orders an hour at peak. What is the recovery point objective (RPO)?

- A.** 15 minutes.
- B.** 1,200 orders.
- C.** 4 hours.
- D.** Zero, because two regions mean data is never lost.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12. [TRANSFER — reversed stem / different variable]

A company must be able to restore its billing system within 2 hours of a failure. Which metric does the 2 hours define?

- A.** Recovery point objective (RPO).
- B.** Mean time between failures (MTBF).
- C.** The uptime percentage in the service-level agreement.
- D.** Recovery time objective (RTO).

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer C Concept 1

A network technician at a 40-person law firm replaces an old hub with a device that learns MAC addresses and forwards each frame only to the port where the destination device is connected. ~~The new device came from a local reseller and has blue status lights.~~ At which OSI layer does it mainly operate?

Target word: OSI layer of the device **Decides:** Forwards frames by MAC address = switch.

A	Camouflage	Physical features do not decide the layer; MAC-based forwarding does.
B	Confusion set	Transport handles TCP/UDP segments and ports.
C	Correct	A switch uses MAC addresses to forward frames at Layer 2.
D	Adjacent category	Layer 3 devices route by IP address, not MAC.

2. Answer B Concept 1 • transfer

Which device makes forwarding decisions using IP addresses at Layer 3?

Target word: device from the layer — reverse **Decides:** IP-based forwarding = router.

A	Confusion set	A hub repeats signals at Layer 1.
B	Correct	Routers forward packets between networks by IP address.
C	Inverted operation	A switch forwards by MAC address at Layer 2.
D	Adjacent category	An access point bridges wireless clients at Layer 2.

3. Answer B Concept 2

A company's video-conference system streams live audio and video to remote staff. ~~The vendor charges per seat, and IT runs it over the main office's fiber connection.~~ Engineers want the lowest delay and can accept an occasional dropped frame. Which transport protocol fits best?

Target word: transport protocol **Decides:** Low delay matters more than guaranteed delivery.

A	Confusion set	Retransmissions add delay that hurts live media.
B	Correct	Connectionless and low-overhead; fits real-time media.
C	Camouflage	The physical link does not decide the transport protocol.
D	Adjacent category	ICMP carries network diagnostics and error messages, not media.

4. Answer A Concept 3

An administrator must let remote staff reach an internal Linux server through an encrypted command-line session. ~~The server has 64 GB of RAM and was installed last spring.~~ Which port should the firewall allow?

Target word: port for the service **Decides:** Encrypted command line = SSH.

A	Correct	SSH uses port 22.
B	Wrong stem number	The RAM figure, not a port.
C	Adjacent category	RDP is a graphical remote desktop, mainly for Windows.
D	Confusion set	Telnet gives a command line but sends everything unencrypted.

5. Answer D Concept 4

A branch office is assigned the network 192.168.10.0/26. ~~The branch has 45 employees and two printers.~~ How many usable host addresses does the subnet provide?

Target word: usable hosts in a /26 **Decides:** 6 host bits: $2^6 - 2$.

A	Dropped element	Forgets to subtract the network and broadcast addresses.
B	Wrong stem number	Adds the employees and printers; the device count does not decide subnet size.
C	Confusion set	That is a /24.
D	Correct	$32 - 26 = 6$ host bits; $2^6 = 64$; minus network and broadcast = 62.

6. Answer C Concept 4 • transfer

A team needs one subnet with at least 100 usable host addresses, using the smallest block that works. Which prefix length should it use?

Target word: solve for the prefix **Decides:** Need 7 host bits: $2^7 - 2 = 126$.

A	Confusion set	A /32 identifies a single host address.
B	Inverted operation	A longer prefix means fewer hosts: only 62.
C	Correct	7 host bits give 126 usable hosts, the smallest block with at least 100.
D	Adjacent category	254 hosts works but is not the smallest block.

7. Answer B Concept 5

A hospital's patient portal went offline for six hours after a flood of traffic from thousands of compromised devices. No records were viewed or changed. ~~The portal was built by an outside vendor in 2019.~~ Which part of the CIA triad was most directly affected?

Target word: CIA element hit **Decides:** Service unusable; data not seen or changed.

A	Confusion set	No records were exposed.
B	Correct	A denial-of-service attack keeps users from reaching the system.
C	Confusion set	No records were altered.
D	Camouflage	The build date does not decide which property the attack hit.

8. Answer C Concept 5 • transfer

Which control mainly protects integrity?

Target word: control for integrity — reverse **Decides:** Detects unauthorized changes.

A	Absolute language	No single control covers all three equally.
B	Adjacent category	Redundancy protects availability.
C	Correct	A changed hash shows the data was altered.
D	Confusion set	Encryption mainly protects confidentiality.

9. Answer A Concept 6

Two offices must send a large volume of backup data to each other every night, quickly, and they have already exchanged a secret key securely. ~~The data is about 2 TB and the link is 1 Gbps.~~ Which approach is most efficient for encrypting the bulk data?

Target word: encryption type for bulk data **Decides:** Shared secret key plus large data = symmetric.

A	Correct	Fast and efficient for large data with a shared key.
B	Camouflage	Link speed does not decide whether data needs protecting.
C	Adjacent category	Much slower; used for key exchange and signatures.
D	Confusion set	Hashing is one-way; it cannot be decrypted.

10. Answer D Concept 7

A new payroll clerk at a 300-employee manufacturer needs to enter hours and run pay reports. Her manager has admin rights to the whole HR system and suggests copying his permissions to save time. ~~The clerk transferred from the company's Chicago plant.~~ What should the administrator do?

Target word: correct access decision **Decides:** Her job needs only payroll entry and reports.

A	Adjacent category	Too broad in scope and too narrow to enter hours.
B	Inverted operation	Gives the most access instead of the least.
C	Camouflage	Her past location does not decide her access needs.
D	Correct	Least privilege.

11. Answer A Concept 8

A retailer's leaders decide that after an outage the online store must be running again within 4 hours, and no more than 15 minutes of orders may be lost. ~~The store runs in two cloud regions and processes 1,200 orders an hour at peak.~~ What is the recovery point objective (RPO)?

Target word: RPO **Decides:** Allowed data loss = 15 minutes.

A	Correct	RPO is the maximum tolerable data loss, measured in time.
B	Wrong stem number	The peak order rate, not the objective.
C	Confusion set	That is the RTO, the restore time.
D	Absolute language	Redundancy lowers risk but does not set the objective at zero.

12. Answer D Concept 8 · transfer

A company must be able to restore its billing system within 2 hours of a failure. Which metric does the 2 hours define?

Target word: name the metric — reverse of RPO **Decides:** Time to restore = RTO.

A	Inverted operation	RPO measures tolerable data loss, not restore time.
B	Adjacent category	MTBF measures reliability between failures.
C	Confusion set	Uptime is a contract measure, not the restore target.
D	Correct	RTO is the target time to restore service.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	1	
Wrong stem number	A figure copied from the prompt that is not the answer.	3	
Adjacent category	The right idea in the wrong bucket.	9	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	4	
Absolute language	Always, never, guarantee — where judgment applies.	2	
Confusion set	Every option is real; the test is exact role-matching.	12	
Camouflage	Answers a stem fact that was never the question.	5	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	OSI layers and devices	Hubs work at Layer 1, switches forward frames by MAC address at Layer 2, and routers forward packets by IP address at Layer 3.	Placing switches at Layer 3 or routers at Layer 2.	“MAC address” (Layer 2) vs. “IP address” (Layer 3).
2	TCP vs. UDP	TCP is connection-oriented and guarantees ordered delivery; UDP is connectionless and faster, trading reliability for low delay.	Choosing TCP for live voice and video because reliable sounds better.	“Lowest delay” and “can tolerate some loss.”
3	Well-known ports	SSH 22, Telnet 23, SMTP 25, DNS 53, HTTP 80, HTTPS 443, RDP 3389.	Picking Telnet (unencrypted) for secure remote shell, or RDP for command line.	“Encrypted command-line session” or a named service.
4	Subnet sizing	Usable hosts = $2^{(32 - \text{prefix})} - 2$; a longer prefix means fewer hosts.	Forgetting to subtract the network and broadcast addresses, or reversing prefix size and host count.	A /n prefix and “how many usable hosts.”
5	CIA triad	Confidentiality keeps data private, integrity keeps it accurate and unaltered, availability keeps systems usable when needed.	Calling every attack a confidentiality breach.	Outage (availability), altered data (integrity), exposed data (confidentiality).
6	Symmetric, asymmetric, hashing	Symmetric encryption (AES) is fast for bulk data; asymmetric (RSA) is slow and used for key exchange and signatures; hashing is one-way and checks integrity.	Using asymmetric encryption for bulk data or calling hashing encryption.	Large data volume plus a shared secret key.
7	Least privilege	Give each user only the access needed for their job, nothing more.	Copying a manager’s rights to save time.	A new role and an offer of broad or copied permissions.
8	RTO vs. RPO	RTO is how fast a system must be restored; RPO is how much data, measured in time, the business can afford to lose.	Swapping RTO and RPO.	“Back up and running within” (RTO) vs. “no more than X minutes of data lost” (RPO).

Course scope (public description): Foundations - Network and Security - Foundations introduces learners to the basic network systems and concepts related to networking technologies. Learners will gain skills in applying network security concepts for business continuity, data access, and confidentiality, and in identifying solutions for compliance with security guidance.